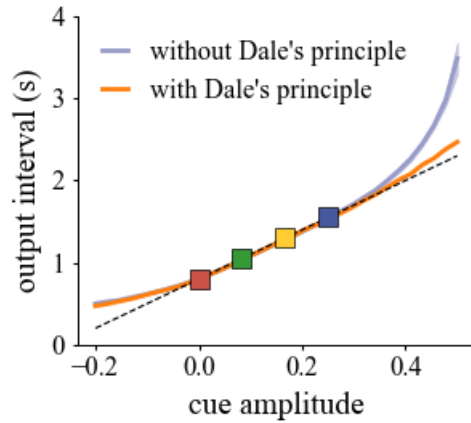
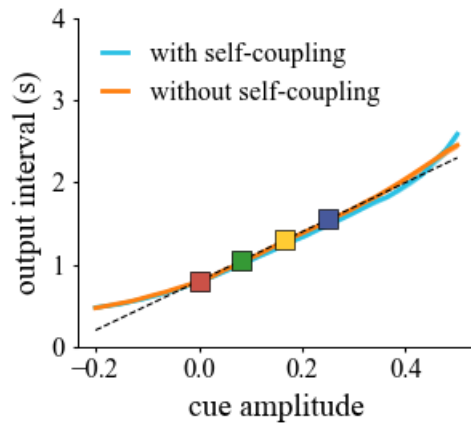

Biologically plausible heavy-tailed connectivity enhances generalizations on cognitive tasks in recurrent neural networks

1



(a)



(b)

Figure 1: Complete ablation of heavy-tailed RNNs for the CSG task: (a) to isolate the effect of the distributional constraint from the sign constraint; (b) to test the role of self-connection.

2

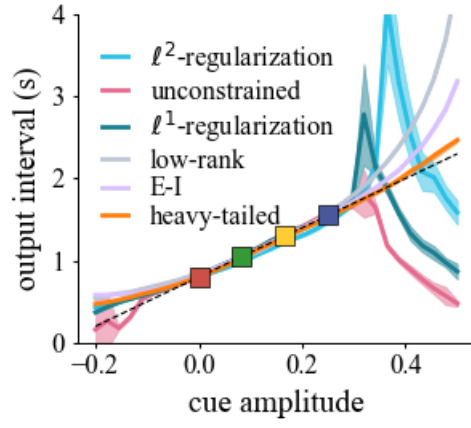


Figure 2: Comparison of the generalization performance of the L^2 -regularized baseline with other RNNs for the CSG task.

3

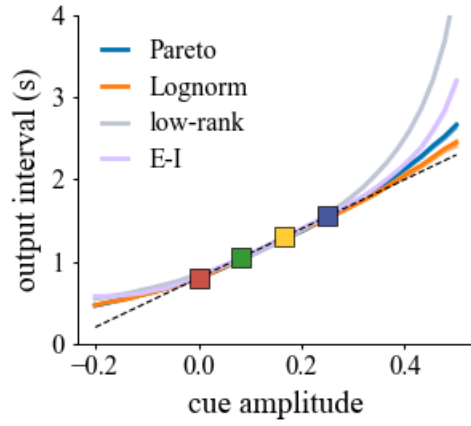
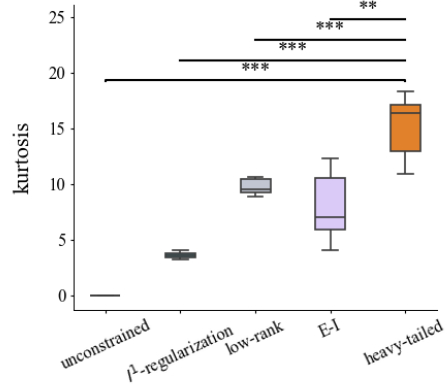
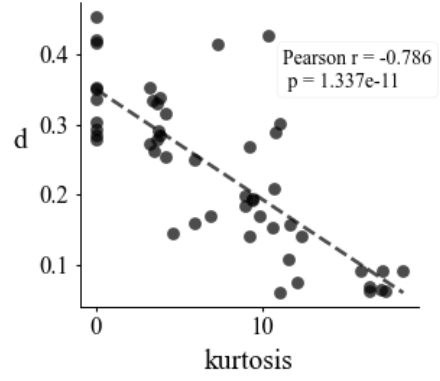


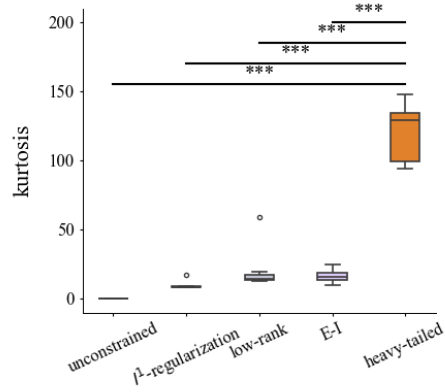
Figure 3: Generalization performance of the trained RNNs with the Pareto-distributed connection for the CSG task.



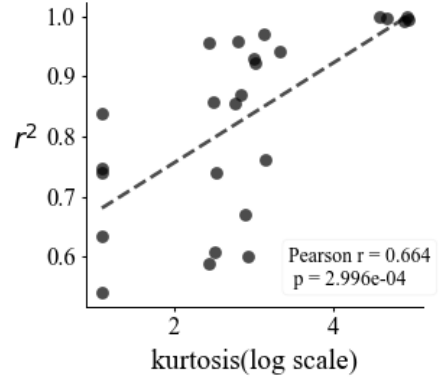
(a)



(b)

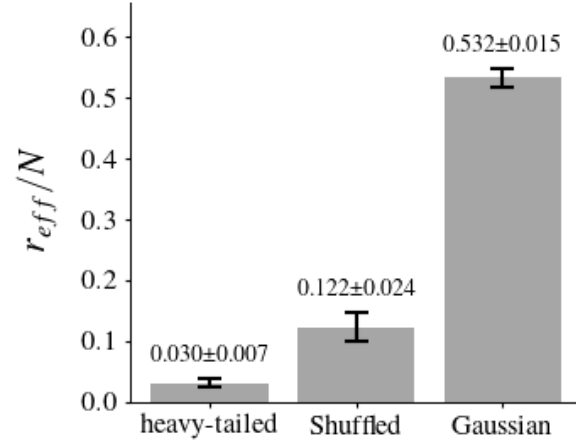


(c)

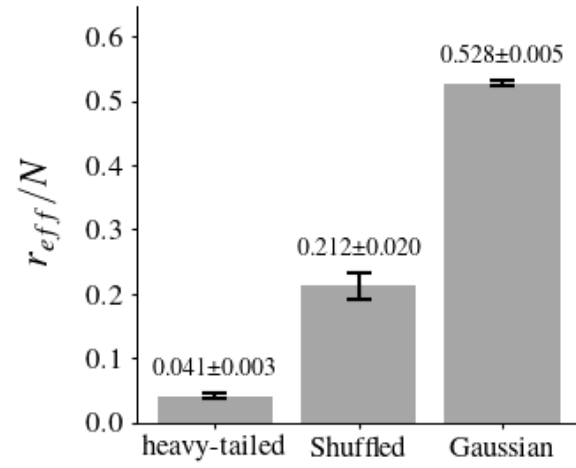


(d)

Figure 4: (a) Kurtosis of different models for the CSG task. (b) Generalization performance versus the excess kurtosis of distributions for the CSG task. (c) Kurtosis of different models for the MWG task. (d) Measurement r^2 of linear representation versus the excess kurtosis for the MWG task.



(a) CSG



(b) MWG

Figure 5: Comparison of the effective ranks across different weight matrices. 'Heavy-tailed' refers to the original weight matrix of the heavy-tailed RNN. 'Shuffled' denotes the heavy-tailed matrix with its elements randomly permuted. 'Gaussian' represents a random matrix whose elements are drawn from a Gaussian distribution with the same mean and variance as the heavy-tailed matrix.